FedMed — Week 1 Working Model & Visual Results Report

Cross-Silo Federated Learning Engine for Privacy-Preserving Healthcare

Week 1 focus: establish a reproducible centralized 3D medical-image segmentation baseline and prepare three isolated hospital node services for the federated phase.

1. Week 1 Objective

The first week establishes the machine-learning foundation before federated training and encryption are introduced. The working pipeline is: MRI-like 3D volume → preprocessing → 3D U-Net → segmentation mask → Dice/IoU evaluation. Three independent hospital services are also prepared on separate local ports so that Week 2 can replace the scaffolding with real Flower clients and FedAvg.

2. System Architecture

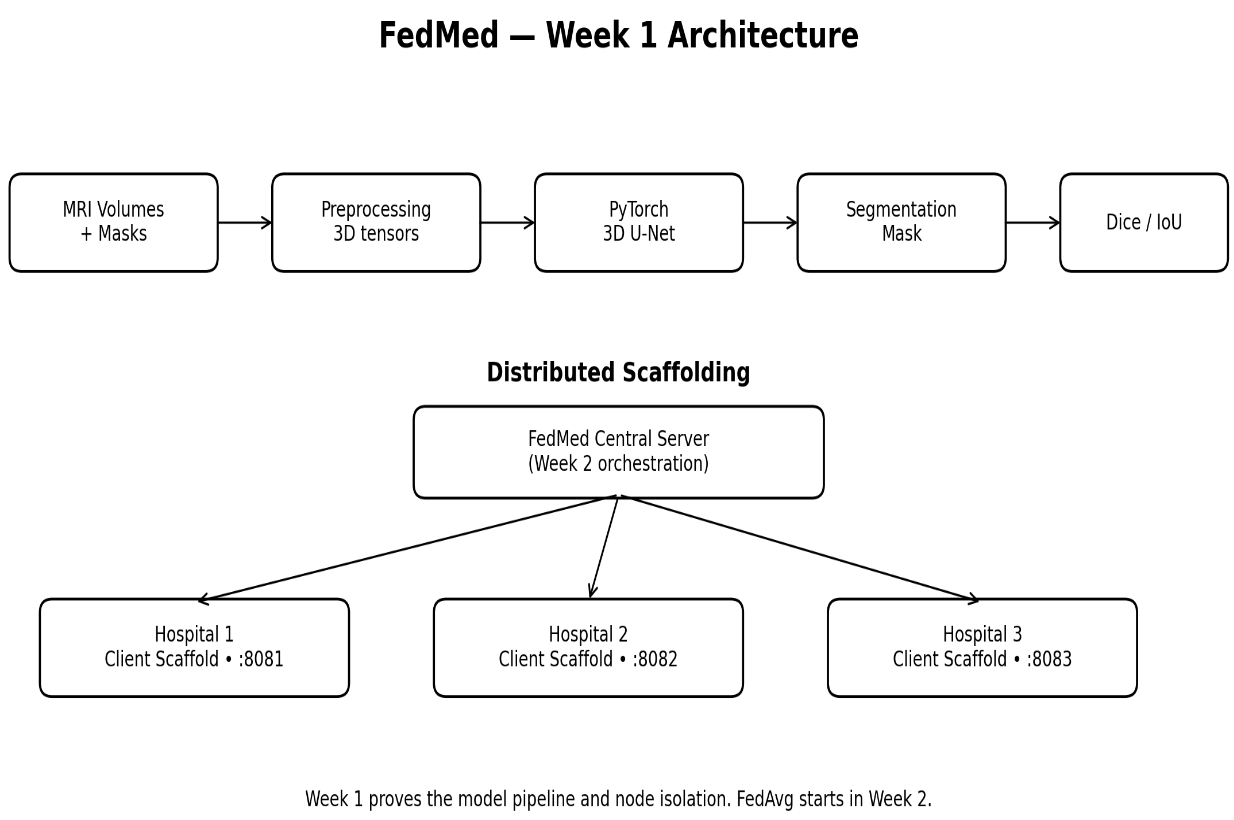

Figure 1. Week 1 architecture: centralized baseline plus three-hospital service scaffolding.

3. Working Model

Model: compact 3D U-Net implemented in PyTorch. It uses encoder blocks to learn volumetric features, a bottleneck for high-level representation, and decoder blocks with skip connections to reconstruct a voxel-level tumor mask.

Input: one-channel 3D volume. **Output:** two-class voxel logits (background and foreground/tumor). **Loss:** cross-entropy. **Optimizer:** Adam. **Evaluation:** foreground Dice and IoU.

4. Reproducible Demo Dataset

For the repository demo, a synthetic MRI-like 3D dataset is generated locally. Each volume contains noisy intensity patterns and a synthetic spherical lesion with a known ground-truth mask. This makes the pipeline runnable without distributing protected clinical data or requiring a large dataset in Git.

Important: synthetic results are engineering/demo results only. They must not be presented as BraTS performance or clinical validation.

5. Segmentation Visual

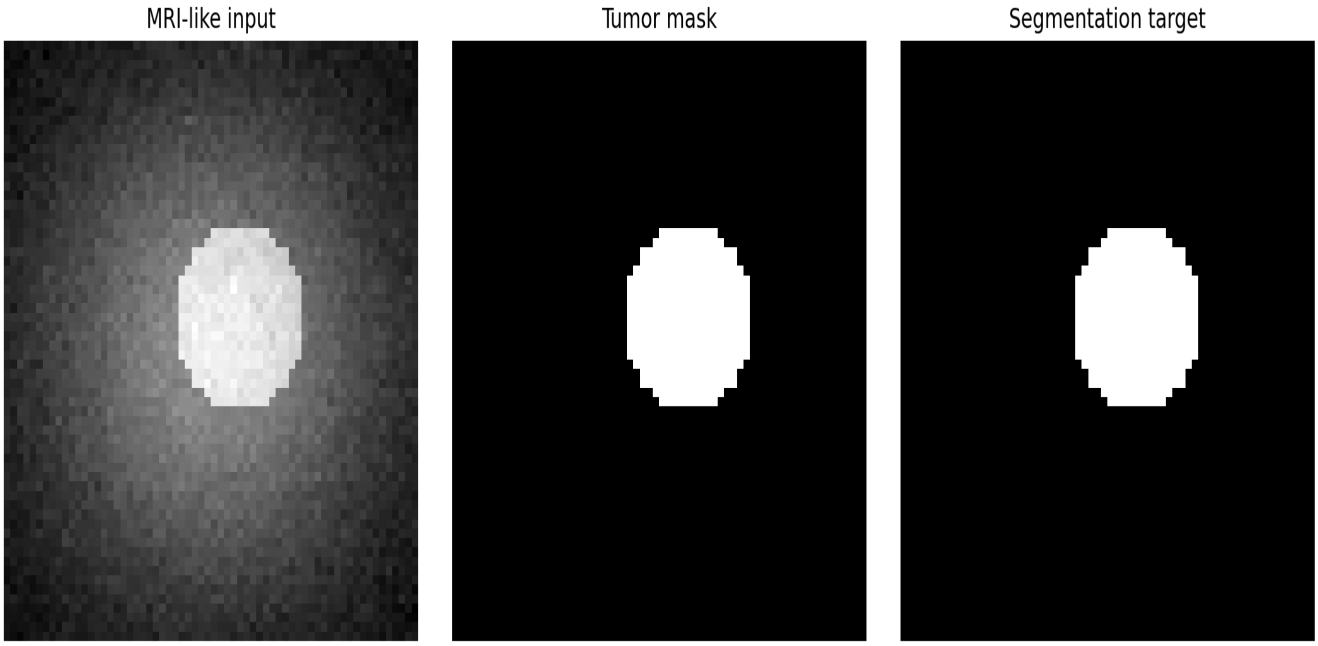

Figure 2. Representative synthetic MRI/segmentation visual included with the Week 1 package. Running the demo script regenerates this asset using the model prediction.

6. Results and Evaluation

No training metrics file is present yet. Run the Week 1 demo/training command to generate actual metrics.

7. Hospital Node Working Model

Three Flask services represent Hospital 1, Hospital 2, and Hospital 3. They are intentionally isolated processes. Each exposes a health endpoint and identifies its hospital role. This is scaffolding rather than federated learning: no patient data is exchanged and no FedAvg aggregation occurs in Week 1.

Node	Port	Purpose
Hospital 1	8081	Independent training-site service
Hospital 2	8082	Independent training-site service
Hospital 3	8083	Independent training-site service

8. How to Run the Working Model

```
pip install -r requirements.txt

python scripts/run_week1_demo.py

python scripts/launch_hospitals.py --hospital 1 --port 8081

python scripts/launch_hospitals.py --hospital 2 --port 8082

python scripts/launch_hospitals.py --hospital 3 --port 8083
```

9. Week 1 Deliverables

Deliverable	Status
3D U-Net implementation	Included
Reproducible 3D synthetic dataset	Included
Training + validation loop	Included
Dice and IoU metrics	Included
Segmentation visualization	Included
Three hospital node services	Included
Flower/FedAvg aggregation	Planned for Week 2
TenSEAL encrypted updates	Planned for Week 3
Differential privacy + React dashboard	Planned for Week 4

10. Engineering Interpretation

Week 1 establishes the baseline needed to answer a key question later: does federated training approach the performance of centralized training while keeping hospital data local? The centralized model therefore acts as the reference point. The hospital services provide the process-level separation needed before introducing Flower, secure transport, encrypted updates, and aggregation.

For a real medical benchmark, the next data step is to integrate a public dataset such as BraTS using appropriate NIfTI loading, normalization, spatial preprocessing, train/validation partitioning, and clinically appropriate evaluation.

Raw medical data should remain outside the Git repository.

11. Git Commit Recommendation

```
git add .
git commit -m "feat: complete FedMed Week 1 baseline and hospital scaffolding"
git push
```